BCA (Sem-III) – Data Structures (301) Core-V

2024

Time : 3 hours

Full Marks : 70

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section–A

Objective type Questions

Compulsory

1. Choose the correct option from the following multiple choice questions : $2 \times 10 = 20$

(a) Process of inserting an element in stack is called :

(i) Create

(ii) Push

(iii) Evaluation

(iv) POP

(b) Which data structure is used to perform level-order traversal on a binary tree?

(i) Stack

(ii) Queue

(iii) Linked List

(iv) Array

(c) What is the primary disadvantage of using an array?

(i) Elements are sequentially accessed

(ii) Index value of an array can be negative

(iii) Wastage of memory space if elements inserted are fewer than allocated size

(iv) All of the above

(d) Which data structure is used for implementing recursion?

(i) Queue

(ii) Stack

(iii) List

(iv) Array

(e) Which data structure follows the Last-In, First-Out (LIFO) principle?

(i) Queue

(ii) Stack

(iii) Linked List

(iv) Tree

(f) Which of the following is NOT a primitive data structure?

(i) Integer

(ii) Float

(iii) Array

(iv) Character

(g) What is the purpose of a hash table?

(i) To store data in a sorted order

(ii) To implement a stack

(iii) To provide fast access to data using a key

(iv) To implement a queue

(h) Which of the following is not the application of stack?

(i) Data Transfer between two asynchronous process

(ii) Compiler Syntax Analyzer

(iii) Tracking of local variables at run time

(iv) A parentheses balancing program

(i) The prefix form of A–B / (C * D ^ E) is?

(i) –A/B*C^DE

(ii) –A/BC*^DE

(iii) –ABCD*^DE

(iv) –/*^ACBDE

(j) What is the value of the postfix expression 6 3 2 4 + – *?

(i) 74

(ii) –18

(iii) 22

(iv) 40

Section–B

Short Answer type Questions

Answer any **four** questions of the following :

5×4=20

2. Explain the different types of data structures.
3. Explain the Selection Sort algorithm with an example.
4. Mention the various applications of the stack data structure.
5. Explain the Binary Search Tree (BST) with its operations.
6. Explain Breadth-First Search (BFS) and Depth-First Search (DFS) algorithms.
7. What is a Queue? Mention its applications.

Section–C

Long Answer type Questions

Answer any **two** questions of the following :

15×2=30

8. What is a Linked List? Briefly explain its types with examples.
9. What is a Pointer? How is dynamic memory allocated using pointers?
10. How is an infix expression converted into a postfix expression?
11. Write an algorithm to insert a node at the beginning of a linked list.

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